Anu Anna Koshy

Rostock, Germany | +49 176 29196719 | anuannakoshy61198@gmail.com | linkedin.com/in/anu-anna-koshy

PROFILE

Research-oriented Electronics and Signal Processing Engineer with an MSc in Information and Communication Technology (FAU Erlangen-Nuremberg), specialising in machine learning for signal processing, statistical signal processing, pattern recognition, and data-driven methods. Over 3 years of research and industry experience in algorithm development, data-driven analysis, and experimental pipeline design. Proficient in Python and MATLAB; experienced in applying deep learning and ML techniques to high-dimensional datasets. Strong foundations in statistical inference, feature extraction, and model-based engineering, with a peer-reviewed publication at PCIM Europe 2025.

EDUCATION

M.Sc. Information and Communication Technology — Oct 2022 – Oct 2025
FAU Erlangen-Nuremberg, Germany — Erlangen, Germany

Grade: 2.0 | *Relevant coursework:* Machine Learning in Signal Processing, Digital & Statistical Signal Processing, Pattern Recognition, Image / Video & Speech Synthesis, Multimedia Technology

B.Tech (Hons) Electronics and Communication Engineering — Aug 2017 – Jun 2021
College of Engineering, Chengannur, India — Kerala, India

Grade: 1.2 | *Relevant coursework:* Digital Signal Processing, Analog & Digital Communication, Wireless Communications, VLSI, Radar Engineering, Microwave & Optical Networks

Achievement: Top 2% of all successful candidates in Electronics & Communication Engineering, 2021

RESEARCH & WORK EXPERIENCE

Research Assistant – Power Electronics & Power Semiconductors — Nov 2025 – Present
University of Rostock — Rostock, Germany

- Conduct experimental research within an academic setting: design and implement hardware systems, develop structured validation and testing workflows, and produce technical documentation aligned with research objectives.
- Apply Python-based data processing, signal analysis, and algorithm development for experimental measurement systems.
- Collaborate across interdisciplinary teams spanning hardware, software, and simulation in a university research environment.

Master Thesis Researcher – TDDB Analysis in 4H-SiC CMOS up to 500 °C — Mar 2025 – Oct 2025
Fraunhofer IISB — Erlangen, Germany

- Designed and executed a data-driven reliability study on 100+ SiC CMOS test structures: built end-to-end analytical pipelines for data preprocessing, feature extraction, statistical modelling (Weibull analysis), and pattern interpretation from large experimental datasets.
- Applied statistical inference and model-based analysis to identify degradation trends and predict device lifetime, achieving ∼20% improvement in prediction accuracy.
- Conducted structured root-cause analysis by correlating high-dimensional stress parameters with observable failure signatures – directly analogous to causal inference in data-driven research settings.
- Peer-reviewed publication co-authored from this work: PCIM Europe 2025 (DOI: 10.30420/566541313).

Analog / Power Semiconductor Product & Test Engineer — Feb 2023 – Sep 2025
Infineon Technologies — Erlangen, Germany

- Automated data analysis pipelines in Python and MATLAB for characterisation of SiC power semiconductor

devices, reducing engineering evaluation time by ∼20%; experience with large dataset processing, parameter extraction, and result visualisation.

- Designed and executed multi-sweep measurement campaigns and performed signal-level analysis and statistical interpretation of device behaviour across a range of operating conditions.
- Applied machine learning concepts and model-based simulation (LTSpice, SIMetrix, PLECS) to evaluate device performance and predict system behaviour.

Research Intern – TDDB & TZDB in SiC CMOS May 2024 – Jul 2024
Fraunhofer IISB Erlangen, Germany

- Investigated degradation mechanisms through experimental data analysis and statistical interpretation; contributed to an estimated ∼10% improvement in device lifetime prediction.

Software Engineer – Oracle Techno Functional Consultant Jul 2021 – Sep 2022
Aspire Systems (India) Pvt. Ltd. Kerala, India

- Developed enterprise software integrations using Oracle Integration Cloud and ERP systems; built scripting and automation solutions for large-scale production environments.

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning for Signal Processing, Deep Learning concepts, Statistical Inference, Pattern Recognition, Data-Driven Analysis, Classification, Regression, Feature Extraction

Signal & Data Processing: Digital & Statistical Signal Processing, Adaptive Filtering, STFT / Fourier Analysis, Spectral Analysis, Weibull Statistical Modelling, High-Dimensional Data Pipelines

Programming: Python (data analysis, automation, pipeline development), MATLAB, C, C++, VBA, PL/SQL

Deep Learning Frameworks: Familiar with PyTorch and TensorFlow for ML model development and experimentation

Simulation & Tools: LTSpice, SIMetrix, PLECS, Origin, KiCad, Eagle

Hardware & Embedded: PCB Design, Mixed-Signal Design, Arduino, Raspberry Pi, VHDL, Verilog, Fixed-Point Implementation

Platforms: Git, Oracle Integration Cloud, Oracle SQL Developer

Languages: English – Business Fluent (IELTS C1, 2022) | German – Intermediate

PUBLICATION

Anu Anna Koshy et al., *"SiC MOSFET Compact Modeling,"* peer-reviewed publication presented at **PCIM Europe 2025**. DOI: 10.30420/566541313

SELECTED PROJECTS

- **Genomic Signal Processing:** Applied digital signal processing methods to genomic sequence data – short-time Fourier transform (STFT) spectral analysis and colour spectrogram visualisation for DNA base pattern recognition. Demonstrates experience in applying signal processing and feature extraction to structured data, directly relevant to computer vision and visual pattern analysis workflows.
- **Automatic Food Quality Analyser:** Embedded sensor system with automated classification logic using Arduino Uno, gas sensors, and LCD interfacing – involves real-time data acquisition, feature-based decision making, and classification pipeline design.
- **Smart Helmet for Alcohol Detection & Accident Avoidance:** Integrated sensing and actuation system combining signal measurement, threshold-based inference, and automated response.